

Introduction to Statistics, Examples

CSU, Chico Math 314

2025-08-28

outline

Rehearse Population/Sample

Sample Analysis

Take Away

References

Population

To which population do the following questions refer?

- ▶ Does smoking cause lung cancer?
- ▶ Do Gmail users receive (to their inbox) less spam than Outlook users?
- ▶ Does LibreOffice crash less often than Microsoft Word?

Sample

What would a random sample look like from the following situations?

- ▶ Does smoking cause lung cancer?
- ▶ Do Gmail users receive (to their inbox) less spam than Outlook users?
- ▶ Does LibreOffice crash less often than Microsoft Word?

Experiments or Observational Study

What would it take to make the following experiment?

- ▶ Does smoking cause lung cancer?
- ▶ Do Gmail users receive (to their inbox) less spam than Outlook users?
- ▶ Does LibreOffice crash less often than Microsoft Word?

outline

Rehearse Population/Sample

Sample Analysis

Take Away

References

Sample Analysis

Suppose we're attempting to measure computer performance. We'll use the unit millions of instructions per second (mips). We'll consider the data set named `speed`, which contains information on various systems mips.

NOTE: The URL in the code must be entered on a single line to work:

```
speed <- read.csv("https://raw.githubusercontent.com/roualdes/data/master/speed.csv")
tail(speed)
```

```
##           system    mips year cores
## 131      AMD FX-8350  97125 2012   1
## 132 Intel Core i7 3770K 106924 2012 1
## 133 Intel Core i7 3630QM 113093 2012 1
## 134 Intel Core i7 4770K 133740 2013 1
## 135 Intel Core i7 5960X 238310 2014 1
## 136   Raspberry Pi 2   4744 2014 1
```

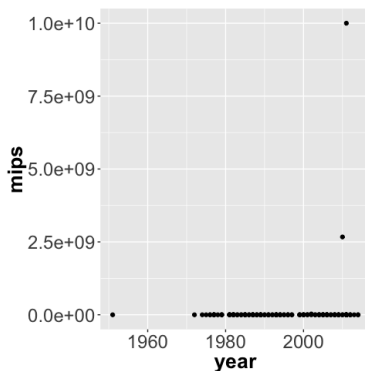
Sample Analysis - First Step

Plot the data! What kind of data do we have? What kind of plot does that warrant?

Sample Analysis - Plotting the Data

Make a scatter plot.

```
suppressMessages(library(ggplot2))  
ggplot(data=speed, aes(year, mips)) +  
  geom_point()
```

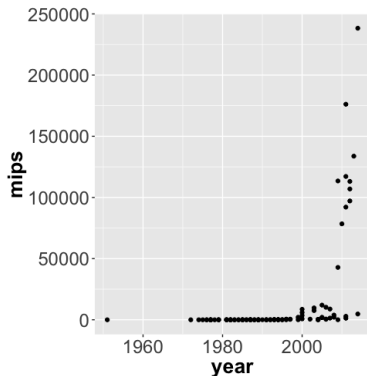


Sample Analysis - Concerns

What's wrong with this plot? Is there a lurking variable we are missing?

Sample Analysis - Corrections

```
suppressMessages(library(dplyr))  
ggplot(data=dplyr::filter(speed, cores == 1),  
       aes(year, mips)) +  
  geom_point()
```



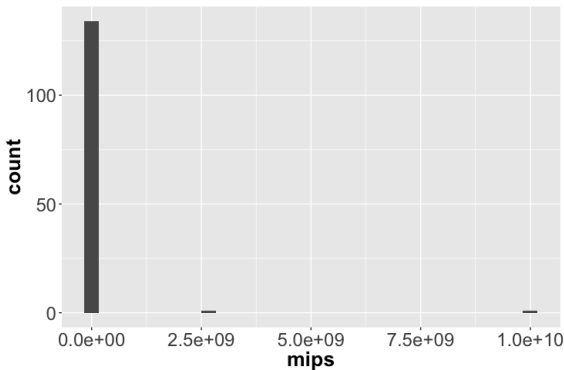
Sample Analysis - Considerations

From the above scatter plot, what kind of skew do we expect mips to have? What is the univariate plot that we should use to look at mips?

Sample Analysis - Histogram

Histogram of mips.

```
ggplot(aes(mips), data=speed) +  
  geom_histogram(bins=31)
```



Sample Analysis - Summary Statistics

Ignoring the variable cores, how can we find the mean and median?

Sample Analysis - Summary Statistics

The functions `mean` and `median`.

```
mean(speed$mips)
```

```
## [1] 93427146
```

```
median(speed$mips)
```

```
## [1] 146.35
```

Sample Analysis - Summary Statistics

Let's also calculate the standard deviation of mips.

```
sd(speed$mips)
## [1] 885866970

sd(filter(speed, cores == 1)$mips)
## [1] 39920.26
```


outline

Rehearse Population/Sample

Sample Analysis

Take Away

References

Take Away

From a little analysis comes little conclusions.

- ▶ What can we conclude with respect to
 - ▶ sample and population?
 - ▶ plots?
 - ▶ mean, median, standard deviation?
 - ▶ What do we think of our simple models?
- ▶ What better questions might we be interested in?

outline

Rehearse Population/Sample

Sample Analysis

Take Away

References

references I

Hadley Wickham and Garrett Grolemund. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. "O'Reilly Media, Inc.", 2016.